PharmAchieve

CHRONIC SPASTICITY

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care: Clinical Care

Providing Care: Distribution

Knowledge & Expertise

Communication & Collaboration

Leadership & Stewardship

Professionalism

LEGEND

ADRs	Adverse Drug Reactions
ALS	Amyotrophic Lateral Sclerosis
CBD	Cannabidiol
CNS	Central Nervous System
CYP	Cytochrome P450
DVT	Deep Vein Thrombosis
FES	Functional Electrical Stimulation
GABA	Gamma-Amino Butyric Acid
LFT	Liver Function Tests
MS	Multiple Sclerosis
N/V	Nausea/Vomiting
S/E	Side Effects
TCA	Tricyclic Antidepressants
THC	Tetrahydrocannabinol

DEFINITIONS AND PATHOPHYSIOLOGY

- Spasticity: velocity-dependent ↑muscle tone in which the muscles involuntary contract, stiffen and tighten
    - Results from injury to motor pathways in the brain or spinal cord
- Chronic spasticity: characterized by the imbalance of excitatory and inhibitory inputs to the motor neurons resulting from injury to the brain and/or spinal cord

Pyramidal upper motor neurons are *not functioning* → Lower motor neurons keep the muscle in the contracted state → The ↑contracted state causes ↑muscle tone, resulting in chronic spasticity

CLINICAL PRESENTATION

- ↑ Muscle tone
- Muscle stiffness and fatigue
- Overactive reflexes
- Involuntary movements which may include spasms or clonus
- ↓ Joint range of movement
- Bone and joint deformities
- Abnormal posture
- Impaired bowel/bladder function
- Significant pain
- Skin breakdown
- Difficulties with care and hygiene
- Contractures (the shortening of muscles leading to joint dysfunction)

RISK FACTORS

- Amyotrophic Lateral Sclerosis (ALS)
- Cerebral Palsy
- Multiple Sclerosis (MS)
- Brain tumor/injury
- Spinal tumor/injury

MEDICAL CAUSES

- Infection
- Pressure sore
- Noxious stimulus
- DVT
- Bladder distention
- Cold weather
- Fatigue
- Seizure activity
- Stress
- Mispositioning

GOALS OF THERAPY

- ✓ Minimize spasms, skin breakdown and pain
- ✓ Improve seating and positioning, gait and functional movements
- ✓ Increase the range of movement of the patient
- ✓ Reduce falls risk

Spasticity should not always be treated because it can worsen a patient’s gait and movement by unmasking the weakness in their limbs.

Only treat when it causes pain and interferes with the daily functions and care of the patient

NON-PHARMACOLOGICAL THERAPIES

- Degree of spasticity varies between patients and treatment should be individualized
- Stretching and joint mobilization exercises can help to relieve the symptoms and ↑ range of movement
- Positioning orthoses may be considered on an individual basis
- Routine use of splints alone for spasticity is not recommended (may be considered for complications, on an individual basis)
- Referrals to physical and occupational therapists can be made to help patients to maximize comfort and range of motion
- Functional electrical stimulation (FES), an adjunct to physiotherapy can be beneficial to selected individuals predominantly affected by upper motor neuron pathologies resulting in a dropped foot

MANAGEMENT OF CHRONIC SPASTICITY

Diagnosed chronic spasticity

Identify and eliminate triggers

Continuing spasticity that interferes with patient's daily functioning and comfort

No → No drug therapy required

Yes →

Generalized spasticity
- Baclofen
- Add or switch to tizanidine
- Add/switch to gabapentin or THC/CBD buccal spray
- Add benzodiazepines for nighttime control of spasms/spasticity
- Baclofen intrathecal pump

Focal spasticity
- Botulinum toxin type A or injections
- Consider permanent selective neurotomy

At any time, consider adding botulinum toxin type A or phenol injections for muscle groups that are severely affected

GENERALIZED SPASTICITY

Baclofen	• Standard initial treatment; safety not established in children <12 years • Intrathecal route available when oral is insufficient or intolerable • Sudden discontinuation can cause seizures, hallucinations and confusion
Tizanidine	• Second-line/combination with baclofen • Safety and efficacy in children <18 years not established
Gabapentin	• Not a Health Canada-approved indication • An alternative when neuropathic pain co-occurs
Benzodiazepines	• Helpful for treating nighttime (nocturnal) spasms
Cannabinoids	• Beneficial add-on; Improves neuropathic pain, sleep disturbance and spasticity. Long term effects are unclear
Dantrolene	• May be considered when baclofen not effective/tolerated • Risk of hepatotoxicity limits its use – LFTs must be strictly monitored

FOCAL SPASTICITY

Local treatment is more effective, avoids the significant CNS side effects associated with systemic therapy and may also benefit patients with spasticity of cerebral origin when other drug therapies are less effective.

Phenol injections

- Block nerves going to specific spastic muscle groups
- Treatment is repeated every 6 months as required
- S/E: pain, dysesthesias and infection

Botulinum toxin type A (e.g. onabotulinumtoxinA or Botox®)

- Better tolerated than phenol
- Blocks the release of acetylcholine at the neuromuscular junction, causing weakness (reduced tone) in the treated muscle and improved passive function
- Effect lasts 3 months

PHARMACOLOGICAL THERAPY

CLASS	SIDE EFFECTS	DRUG INTERACTIONS
GABA-B agonist: Baclofen	Sedation/weakness, nausea, dizziness, lowered seizure threshold; caution in patients with brain injuries; toxic encephalopathy in renal impairment	TCAs, opioids, benzodiazepines, anti-hypertensives, levodopa/carbidopa
GABA Derivatives: Gabapentin	Sedation, dizziness, fatigue, weight gain	Antacids may ↓ absorption of gabapentin (separate by at least 2 hours) CNS depressants
Alpha$_2$-adrenergic agonists: Tizanidine	Dry mouth, sedation/dizziness, hallucinations, hepatotoxicity (monitor LFTs); caution in brain injured patients	Antihypertensives, TCAs, CYP1A2 inhibitors (ciprofloxacin), oral contraceptives
Benzodiazepines: Clonazepam, Diazepam	Sedation, ataxia, weakness, dependence	Potentiates effects of other CNS depressants including alcohol
Botulinum toxins: OnabotulinumtoxinA, IncobotulinumtoxinA, AbobotulinumtoxinA	Hypertonia, ecchymosis, muscular weakness, injection site pain, fever, headache	Drugs interfering with neuromuscular transmission, aminoglycoside antibiotics, neuromuscular blocking agents (e.g. anticholinesterases, quinidine)
Cannabinoids: Delta-9-THC/ cannabidiol	Fatigue, nausea, diarrhea, vertigo, dry mouth	CNS depressants, potent CYP3A4 inhibitors

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	**Degree of Change**	**Timeframe**
Spasticity	Improved	Hours to days

SAFETY ENDPOINTS		
Parameter	**Degree of Change**	**Timeframe**
ADRs from medication therapy (sedation, falls, etc.)	Prevent	Duration of therapy
Recurrent symptoms	Prevent	

TAKEAWAY

Chronic spasticity can limit joint range of motion and cause contractures leading to significant pain, skin breakdown, and difficulties with care and hygiene

Spasticity may be treated when pain interferes with the patient's care and activities of daily living

Daily stretching and exercise can help to relieve symptoms and increase range of motion

Physical therapy focuses on gross motor skills, including range of motion, strength and stability. Occupational therapy focuses on functional and self-help skills.

Generalized spasticity is initially managed with baclofen. Alternative or add-on therapy may include tizanidine, gabapentin, THC/CBD buccal spray, or benzodiazepines

Focal spasticity may benefit from local treatment e.g. phenol injections, botulinum toxin type A

CASE STUDY

JP is a 45-year-old male who presents to your pharmacy following an appointment with his neurologist. Over the past 6 months, he has experienced persistent muscle stiffness, occasional muscle cramps, fatigue, and spasms in both lower limbs at night. He reports that his symptoms interfere with his sleep on most nights of the week and that he no longer feels mobile. His past medical history is significant for hypertension (diagnosed 5 years ago), major depressive disorder (diagnosed 2 years ago), and relapse-remitting multiple sclerosis (diagnosed 8 years ago). His current medications include lisinopril 10 mg PO daily, sertraline 50 mg PO daily, teriflunomide 7 mg PO daily, and vitamin D 1000 units PO daily. JP works primarily at home as a graphic designer and lives independently. JP tells you that he drinks 1-2 glasses of alcohol daily, and he smokes approximately 5 cigarettes daily. Following a diagnosis of chronic spasticity from his neurologist, he tried daily stretching and physiotherapy to improve his range of motion but experienced little improvement. His neurologist would like your assistance in recommending a suitable management plan for JP.

CASE STUDY - QUESTIONS

1. Which of the following agents is most appropriate to recommend as initial therapy?
   a) No pharmacological therapy should be initiated at this time
   b) Gabapentin 100 mg PO TID
   c) Aqueous phenol IM Q6 months
   d) Baclofen 5 mg PO TID

2. Which of the following counselling points is true when counselling on the chosen therapy?
   a) Avoid alcohol as it may worsen drowsiness
   b) Stop the medication immediately if you feel too tired or weak
   c) Take the medication PRN only when you feel spasms
   d) You may discontinue the medication immediately once your spasms resolve

3. Which of the following adverse effects is most important to monitor for?
   a) Hypertension
   b) Hepatotoxicity
   c) Sedation and muscle weakness
   d) Gingival hyperplasia

4. Four weeks after the dose of baclofen has been optimized, JP reports partial symptom relief but significant drowsiness and continued fatigue which affect his daily activities. What is the most appropriate next step?
   a) Discontinue the medication immediately and switch to tizanidine
   b) Add diazepam PRN at bedtime
   c) Reduce the dose and add tizanidine cautiously
   d) Switch to onabotulinumtoxinA

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CHANGE LOG

July 2025

- Formatting, spelling, grammar changes throughout
- Slide 3: Updated legend
- Slide 5: Added definition of contractures
- Slide 13: Changed GABA Derivative to GABA-B agonist for baclofen
- Slide 15: Added takeaway
- Slides 16, 17: Added case study and questions
- Slide 18: Updated references

November 2025

- Formatting edits made; no content changes

December 2025

- Slide 13: Removed phenytoin interaction with tizanidine, added oral contraceptives

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 9: Updated recommendations for clarity
- Slide 11: Added information on dantrolene
- Slide 18: Added references